PBTA Analysis – Summary Report

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1 Summary

Questions like what cancer group is interesting or what is interesting in the data are challenging questions and could be tackled in a few ways. A simple yet powerful approach would be to tackle groups that have strong statistics to work on (number of samples and so on). Analysing these groups would be reproducible among other groups and can reveal novel outcomes. Moreover, Brlek et al. describe multiple ways to interactively use cBioportal to reveal molecular mechanisms in cancer (Brlek et al. [2021]).

Nevertheless, to answer these questions, I conducted a systematic unsupervised approach where I assumed no prior knowledge or bias on the data. I was trying to answer two main questions: Which cancer group is interesting, and which variable significantly differs across groups in the data? To do so, I visualised and statistically questioned the clinical data by asking whether AGE, SEX, TF, TP, RACE, and PREDISPOSITION differ in cancer groups and are related to death/event status. I evaluated 343 combinations and found that HGG, DMG, LGG, ATRT, Chordoma, and Schwannoma are the most interesting cancer groups. AGE differs the most across all tests, and has the most powerful effect in the context of survival.

2 Methods

2.1 Data processing

I processed the clinical metadata of the PBTA RNA study, which includes two tables: patient-level and sample-level tables with 2870 and 4312 samples, respectively. I merged the two tables on the patient ID, which resulted in 4074 samples (2701 unique patients) with a known CG. To analyse the data, I converted the OS and EFS status values into labels and event flags, merged missing RACE and ETHINICITY values into "Unknown", mapped the cancer PREDISPOSITION into "No predisposition"/"Unknown", and merged blank or "To be classified" molecular subtypes into "Unclassified". For the AGE distribution, I detected outliers with the Tukey IQR rule (AGE $>$ Q3 + 1.5*IQR = 27.125 yr), which flagged 62 patients.

2.1.1 Statistic analysis

To evaluate the data statistically, I computed the % of missing values in each column to identify noisy variables. To compare CGs, I tested every CG with n $>=$ 20 samples (excluding 33 CGs): SEX was tested against a 50:50 baseline with a binomial test, while PREDISPOSITION and RACE compositions were tested with Chi-squared tests (Fisher exact when the expected counts were $<$ 5), reporting Cramer's V as the effect size. For the numeric variables AGE, TUMOR_FRACTION (TF), and TUMOR_PLIDY (TP), I used a Kruskal-Wallis test followed by per-group Mann-Whitney tests (each CG vs all the others). I corrected the p-values within each family of tests using the Benjamini-Hochberg FDR method and considered q $<$ 0.05 as significant, always reporting the raw p next to the adjusted q. Finally, I computed Spearman and

Pearson correlations between AGE, TF and TP. I ordered the CGs by similar race composition using hierarchical clustering.

2.1.2 Survival analysis

To test the effect of variables on survival, I used time-to-event analyses on the OS and EFS endpoints. I plotted Kaplan–Meier (KM) curves per CG ($n \geq 20$) and each group's survival was compared against all others combined with log-rank tests (multiple-testing correction as above). Each variable was then tested against survival status (Mann–Whitney U for AGE, TF, and TP; Chi-squared for SEX and predisposition), and numeric variables were dichotomised at their median to test KM separation with log-rank tests. Finally, I fitted a multivariate Cox proportional-hazards model with AGE, SEX, TUMOR_FRACTION, and TUMOR_PLOIDY as covariates, stratified by cancer group, on complete cases, reporting Wald p-values, hazard ratios with 95% confidence intervals, and Schoenfeld residuals to check the proportional-hazards assumption. For AGE deciles, I divided the CG into subgroups based on the AGE values in 10% increments.

Analyses were implemented in Python (scipy, lifelines, pandas, statsmodels) as executed notebooks. Source code can be found at: `https://github.com/AlonLuboshitz/menow_home_ass`

3 Results

I conducted four analyses as follows: metadata analysis, Cancer group variable enrichment, survival effects, and Unsupervised clustering. The full notebooks that support these analyses can be found under notebooks/*executed.ipynb files. To summarise the supportive claims I produced an HTML notebook (only outputs) for all the sections combined. The figures referenced in this text refer to the HTML notebook. I further applied unsupervised clustering techniques, but due to the time limit, I wasn't able to properly interpret the results. You are welcome to check the clinical_mulltivar_hidden_strcture_analysis_executed interactive notebook.

3.1 Metadata and clinical analysis

I conducted a simple visualisation and summarisation of the clinical and demographic data, including statistics and histograms. I noticed that event-free survival (EFS) and overall survival (OS) months variables have 28% Null values (Figure 1A). In total 65.3% have a living status via the OS variable, whereas for the EFS variable, half the patients exhibit no event and the other half are split between different events (Figure 1B, C). This demonstrates the value of investigating the EFS rather than OS, where **EFS provides additional information** (Sternberg et al. [2025]) Furthermore, the age of diagnosis (AGE) right-skewed histogram suggests outliers or misleading data points, as this is a kids dataset (Figure 1D). A further inspection revealed that most of these data points (53/62) belong to oligodendroglioma patients (Figure 1E). A quick web search suggested that this cancer is an adult cancer, and therefore **removing these data points for subsequent analysis should be considered**.

Investigating the samples dataset demonstrated 20-40% Null values in Tumor fraction (TF) and ploidy (TP), RNA selection, and matching ID samples (Figure 2A). Among the 55 cancer groups, Low-grade glioma (LGG) has the most cases (862), followed by HGG, MB, DMG, and Ependymoma. 7 CGs (ATRT, Ganglioglioma, AC, etc.) have 70-200 cases, and 45 groups have fewer than 70 cases (Figure 2B). The molecular subtype information aligns well with the CGs, despite 13 molecular subtypes belonging to 2 or more distinct CGs (Figure 2C). This suggests either wrong labelling of the submolecular type or that some subtypes belong to multiple CGs.

3.2 Cross datasets analysis

Next, to reveal differences between cancer groups, I evaluated whether certain variables are enriched across different cancer groups. To do so, I evaluated the categorical variables SEX, PREDISPOSITION, and RACE using Binomial, Chi-squared and Fisher exact tests. For the numeric variables AGE, TF, and TP, I used Kruskal-Wallis and per-group Mann-Whitney tests across CGs.

I found that for 6 CGs - LGG, Ependymoma, MB, Ganglioglioma, AC, and ES the male-to-female distribution is significantly different (Figure 3A) compared to the 50:50 baseline. Furthermore, among patients with predispositions (9%), 5 CGs - ATRT, HGG, LGG, Neurofibroma, and Schwannoma have different predisposition composition (Figure 3B). Lastly, 9 CGs showed differences in race composition compared to the overall race composition in the dataset (Figure 3C). These findings suggest that **sex, race and predisposition could influence cancer**. Lastly, as demonstrated in (Figure 3C), some CGs show similar predisposition compositions, suggesting a relation between these CGs. A further inspection of these CGs could reveal the connection. Next, I demonstrated that **14, 6, and 4 CGs are significantly different by AGE, TF, and TP variables** (Figure 3D, E, F). Investigating their Spearman/Pearson correlations (Figure 3G, H) yielded significant p values (5/6) with no observed trends, possibly due to the high sample size. Applying numeric transformations to these variables could reveal stronger concurrence.

3.2.1 Survival analysis

To test differences in survival patterns globally and across CGs, I evaluated all the previously analysed variables and tested their effect on OS and EFS. First, I plotted the KM curves for the top CGs and compared each group's survival rates against the rest of the groups (Figure 4A, B). I demonstrated that **15 and 16 CGs have statistically different KM plots** for OS and EFS, respectively. Moreover, in both plots, Diffuse-midline glioma (DMG) has the worst outcomes. Atypical Teratoid Rhabdoid Tumour (ATRT) and High-grade glioma are similar and next after. The Dysembryoplastic neuroepithelial tumour (DNT) has no deaths, though it exhibits events. Next, I evaluated whether each variable differs between survival states using the Mann-Whitney test globally. I found that all variables except TP differentiated the OS and EFS outcomes (Figure 4C, D), respectively. This strengthens the informative value of these variables to affect outcomes. To estimate the global difference for these variables on time to event, I dichotomised the variables by their median and categorical values. Investigating the KM plots, I found that TF, SEX, and PREDISPOSITION differ in OS rates (Figure 4E), whereas AGE and SEX differ EFS rates (Figure 4F). Interestingly, higher TF values have a lower death rate (Figure 4g1). Contradictingly, higher TF values have a higher risk for time to event (Figure 4g2). This can be explained by the fact that 600 samples with TF = 1 have a lower death rate, but at high TF ratios (0.7–0.9) there is a high death rate (Figure 4g3). The TF = 1 might bias the analysis, and **when excluding these samples the MW evaluation shifts** (d sign flips) and correlates with the KM plot (Figure 4g4).

Next, I evaluated if the variables differ in the OS and EFS outcomes inside each CG. To do so, for each group, I dichotomised the samples based on the evaluated variables and compared the KM plots. I demonstrated that AGE is significant in Chordoma, Diffuse hemispheric glioma (DHG), HGG, MB in both OS and EFS and Embryonal tumour with multilayer rosettes, Low-grade glioma in EFS and Schwannoma in OS, **making AGE variable highly valuable**. The SEX variable also demonstrates high differential capabilities, whereas TF dont hold any significant value (after FDR correction). Lastly, the TP variable is significant only for Schwannoma EFS (Table 1). To further evaluate the AGE power to differentiate patients inside CG, I divided the AGE variable into 10% increasing ratios and evaluated the differences in the OS and EFS outcomes **rates**. From 22 GCs, 12 and 17 CGs were differentiated by the AGE deciles, a major increment compared to the median split 2. I concluded that variable effects and co-occurrence on outcome rates should be investigated in a more sophisticated way, where there is no need to split groups based on variable cut-offs.

Table 1: Per-cancer-group survival differences for the OS and EFS outcomes. Each variable is shown with an outcome suffix (AGE-OS, AGE-EFS, TF-OS, TF-EFS, TP-OS, TP-EFS, SEX-OS, SEX-EFS); cells show the BH-FDR q-value computed within each cancer group; bold = significant (q<0.05); '–' = not tested (group has <20 samples with complete data for that variable).

Cancer group	AGE-OS	AGE-EFS	TF-OS	TF-EFS	TP-OS	TP-EFS	SEX-OS	SEX-EFS
Adamantinomatous Craniopharyngioma	0.7220	0.7220	–	–	0.7220	0.7220	0.7220	0.9720
Atypical Teratoid Rhabdoid Tumor	0.7110	0.9462	0.9462	0.9462	0.3687	0.9462	**0.0309**	0.1226
CNS Embryonal tumor	0.2610	0.7404	–	–	–	–	0.7404	0.3709
Chordoma	**0.0064**	**0.0273**	–	–	–	–	**0.0064**	**0.0317**
Choroid plexus tumor	0.0780	0.8546	0.6965	0.6965	0.6965	0.2668	0.6965	0.8546
Diffuse hemispheric glioma	**0.0018**	**0.0031**	–	–	–	–	**0.0259**	0.1896
Diffuse midline glioma	0.8750	0.5933	0.5933	0.8973	0.8750	0.8750	0.6557	**0.0473**
Dysembryoplastic neuroepithelial tumor	1.0000	0.7066	1.0000	0.6987	1.0000	1.0000	1.0000	0.2808
Embryonal tumor with multilayer rosettes	0.7052	**0.0008**	–	–	–	–	0.1132	0.1627
Ependymoma	0.8597	0.0624	0.5378	0.8668	0.8597	0.8523	**0.0032**	0.1483
Ewing sarcoma	0.5709	0.5709	0.9363	0.9745	0.6485	0.9745	0.9745	0.5709
Ganglioglioma	1.0000	0.9708	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Glial-neuronal tumor NOS	0.1738	0.6443	0.8048	0.9979	0.9979	0.4834	0.1709	0.3718
High-grade glioma	**0.0024**	**0.0002**	0.6159	0.4757	0.6159	0.7937	0.5623	**0.0372**
Low-grade glioma	0.4608	**<0.0001**	0.0759	0.8336	0.4039	0.8153	0.0759	0.8153
Medulloblastoma	**0.0077**	**0.0116**	0.1373	0.5288	0.5215	0.5215	0.4516	0.5215
Meningioma	0.3753	0.6245	0.9434	0.6245	0.4274	0.9434	0.0664	0.8028
Neuroblastoma	0.9176	0.9176	–	–	–	–	0.3820	0.3820
Neurofibroma/Plexiform	0.8280	0.9503	–	–	0.9080	0.8280	0.6070	0.8280
Pineoblastoma	0.8653	0.0873	–	–	–	–	0.3522	0.9153
Sarcoma	0.7921	0.5891	0.3202	0.5332	0.8837	0.5891	0.9840	0.8837
Schwannoma	**0.0429**	0.5888	0.3319	0.2318	0.1092	**0.0156**	0.8870	0.5888
N significant (q<0.05)	5	6	0	0	0	1	4	3

Table 2: Cancer groups with significant age decile survival differences (multi-group log-rank test per group; BH-FDR q-value adjusted across cancer groups within each phase; three significance tiers).

OS (12/22 significant):

- **q<0.001:** High-grade glioma, Diffuse midline glioma, Choroid plexus tumor, Low-grade glioma, Medulloblastoma, Ewing sarcoma
- **q<0.01:** Diffuse hemispheric glioma, Atypical Teratoid Rhabdoid Tumor
- **q<0.05:** Glial-neuronal tumor NOS, Sarcoma, CNS Embryonal tumor, Chordoma

EFS (17/22 significant):

- **q<0.001:** Adamantinomatous Craniopharyngioma, Diffuse midline glioma, High-grade glioma, Low-grade glioma, Medulloblastoma, Meningioma, Ependymoma
- **q<0.01:** Diffuse hemispheric glioma, Choroid plexus tumor, Embryonal tumor with multilayer rosettes, Schwannoma, Glial-neuronal tumor NOS
- **q<0.05:** Chordoma, Atypical Teratoid Rhabdoid Tumor, Neurofibroma/Plexiform, Dysembryoplastic neuroepithelial tumor, Pineoblastoma

To evaluate the combination of the variables together on time-to-event and disentangle the dependency on variable cut-offs, I applied a Cox-hazard model on the global dataset. For OS, only the AGE variable is significant with 95% CI and HR of (0.95-0.98, 0.97, Figure 4H), meaning

higher age decreases the hazard by 3%. For EFS, both AGE and TP are significant with CIs and HRs of (0.95-0.97, 0.96 and 1.02-1.18, 1.1, respectively, Figure 4I), meaning **higher age decreases the hazard by 3% and higher TP increases the hazard by 9%**. Lastly, I applied a univariate Cox-PH model to every CG and demonstrated that for the OS outcome, in GNT tumour higher AGE increases the risk whereas in DMG, HGG, and MB CGs it decreases the risk (Figure 4J). Moreover, in both OS and EFS outcomes, all variables except TP in OS have a significant effect on risk (Figure 4K). Nevertheless, compared to the previous analysis, **the variables demonstrate a weaker effect** - the number of CGs with significant effect is much lower. This might suggest weaker/modest effects, though per-cancer-group sample sizes (few events), proportional-hazards assumption, and correlated features limit the detectability of weaker effects.

4 Conclusions

Taken together, the analyses highlight **High-grade glioma (HGG)** as the most interesting cancer group: it is the second-largest cohort (512 samples) with one of the highest mortality rates (∼80% OS death), it is differentiated from other groups by age at diagnosis, Li–FraumenI predisposition, and tumor fraction, and **age at diagnosis separates outcomes in both OS and EFS** (multivariate HR $\approx$ 0.95–0.97 per year, with significant age-decile separation in both endpoints). HGG is not the only group worth pursuing. **Diffuse midline glioma (DMG)** is the most clinically urgent target with the highest death rates in the cohort (96.7% OS death, 97.8% EFS events). Nevertheless, the analysis revealed that DMG is homogeneous at the clinical level (in the within-group survival tests, AGE, TF, and TP all failed to separate DMG patients after FDR correction, $q > 0.05$ on both endpoints); Consequently, any separation for these patients can point towards clinically meaningful subgroups. Furthermore, **Low-grade glioma (LGG)**, the largest cohort (862 samples), is differentiated by the most variables (sex, NF-1 predisposition, race, tumour fraction, and ploidy) and, with only 8% OS death but strong age-driven EFS separation, is the natural system for studying progression rather than death. **ATRT** stands out biologically as the youngest-age group, enriched for rhabdoid predispositions and carrying high mortality. Another two small groups create an interest: **Chordoma** ($n \approx 26$; 72% female, significant in all four AGE/SEX survival tests) and **Schwannoma** (NF-2 enriched, older age, and the only group where tumour ploidy affects EFS)

Across the cohort, **AGE remains the most informative variable** both between groups (14/26 CGs, largest effect size) and within groups (5 OS + 6 EFS CGs at the median split, rising to 12 + 17 at age deciles), and it is the only variable robust in the multivariate Cox model for OS. Tumour fraction and ploidy warrant caution: TF's apparent protective effect reverses once the TF = 1 samples are excluded, and TP surfaces only in the EFS multivariate model.

5 Future directions

Future work should (i) remove the adult-age oligodendroglioma outliers before modelling AGE, (ii) apply numeric transformations to the correlated AGE/TF/TP variables to strengthen their association signals, and (iii) follow the clinical subgroups identified here (HGG in particular, alongside DMG and LGG) into the mRNA-expression analysis to reveal the biological pathways behind the age-dependent survival differences.

References

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